

Proposed AI Assurance Case Standard

A candidate-created discussion artifact for discovery - not a statement of Westinghouse policy or current internal controls.

Purpose

Create one portable evidence contract for AI initiatives so strategy, data, technology, business ownership, independent controls, and adoption can make faster, more consistent decisions without diluting authority.

Minimum assurance-case record

Dimension	Required questions	Evidence examples
Mission & boundary	What decision or workflow changes? What may the system not do? Who owns the outcome?	Use-case charter, prohibited uses, process map, consequence classification, accountable owner.
Data pedigree	What data is used? Who owns it? Is it fit, lawful, secure, current, and traceable?	Catalog/lineage, quality tests, access, retention, consent/purpose, representativeness, source status.
Model evidence	How does it behave across intended and adverse conditions? What changes require revalidation?	Evaluation plan/results, challenge sets, uncertainty, bias, robustness, drift, version and change records.
Human authority	Who reviews, decides, overrides, appeals, escalates, or stops the system?	Decision-rights record, user workflow, override/appeal path, training, competency and independence requirements.
Security & compliance	Which cybersecurity, privacy, legal, quality, export, records, regulatory, and customer controls apply?	Threat model, access/logging, privacy assessment, control mapping, review decisions, incident/recovery plan.
Outcome telemetry	How will we know the workflow is useful, safe, adopted, and still within its operating boundary?	Business/operating measures, use and correction patterns, exceptions, incidents, overrides, drift, benefit evidence.

Design principles

Evidence before scale Production posture is earned through a use-case-specific record, not model novelty.	Authority before automation Human decision rights are explicit before the AI is embedded in work.	Reuse before repetition Approved patterns, tests, and controls should compound across use cases.
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Operating postures and review triggers

Posture	Intended use	Triggers to move, hold, or escalate
Explore	Bounded research, sandbox, discovery, feasibility.	Move when outcome/owner/data are qualified; hold for prohibited data/use, no owner, or unclear consequence.
Assist	Human-authoritative recommendations, drafting, retrieval, analysis.	Move when grounding/evaluation, user workflow, source visibility, training, and correction path are adequate.
Control	Production workflow integrated with monitoring and recovery.	Requires validation, release/change control, security/privacy, telemetry, rollback, incident and exception handling.
Independent review	High-consequence, regulated, safety, employment, or other sensitive decision support.	Requires independent qualified authority, formal assurance, challenge testing, records, strict change control, and explicit stop rights.

Exception record

Every exception should identify the requirement, rationale, accountable approver, affected use case and consequence, compensating control, evidence, monitoring, expiration/review date, and closure decision. Exceptions are portfolio signals: repeated exceptions indicate a missing platform capability, weak standard, or unresolved operating tension.

Portfolio decision record

Fund / pilot Outcome, owner, readiness, consequence, evidence plan, adoption burden, and learning value are explicit.	Scale / standardize Operating evidence shows value, control effectiveness, adoption, recoverability, and a reusable pattern.
Remediate / hold Evidence, data, authority, control, adoption, or operating boundary remains insufficient.	Stop The outcome is weak, risk cannot be responsibly bounded, ownership is absent, or better alternatives exist.

Executive scorecard definitions

- **Portfolio:** work by posture and decision; benefit realization; reuse; stop/remediation decisions.
 - **Flow:** time to qualified pilot; review and exception aging; rework caused by late evidence.
 - **Assurance:** lineage/evidence completeness; validation defects; incidents; recovery and closure.
 - **Authority:** unresolved owners; override, appeal, and stop-path behavior.
 - **Adoption:** workflow use, training, repeated use, corrections, and responsible user confidence.
 - **Learning:** reusable patterns, retired controls, updated standards, and new data/platform capabilities.
- Targets should be set only after baseline definitions, source systems, units, owners, and decision use are validated.